When we merge two partitions, we will set one tree root as the child of the root of the other tree: see figure 5.8 to get an illustrated example. This is a huge improvement over the naïve solution, because we won't have to update the partition map for any of the other elements in the partitions merged. Instead, each node in the tree will maintain a link to its parent (we don't need to save links to children, as they are of no use in this case).

The roots of the tree, as mentioned, uniquely identify each partition. So, when we need to find out which partition an element belongs to, we just retrieve the tree node it's pointing to, and walk up to the root of its tree. In method `areDisjoint`, if we do the same for both elements and then compare the roots found, we can easily see if two elements belong to the same partition (if and only if the two roots are equal).

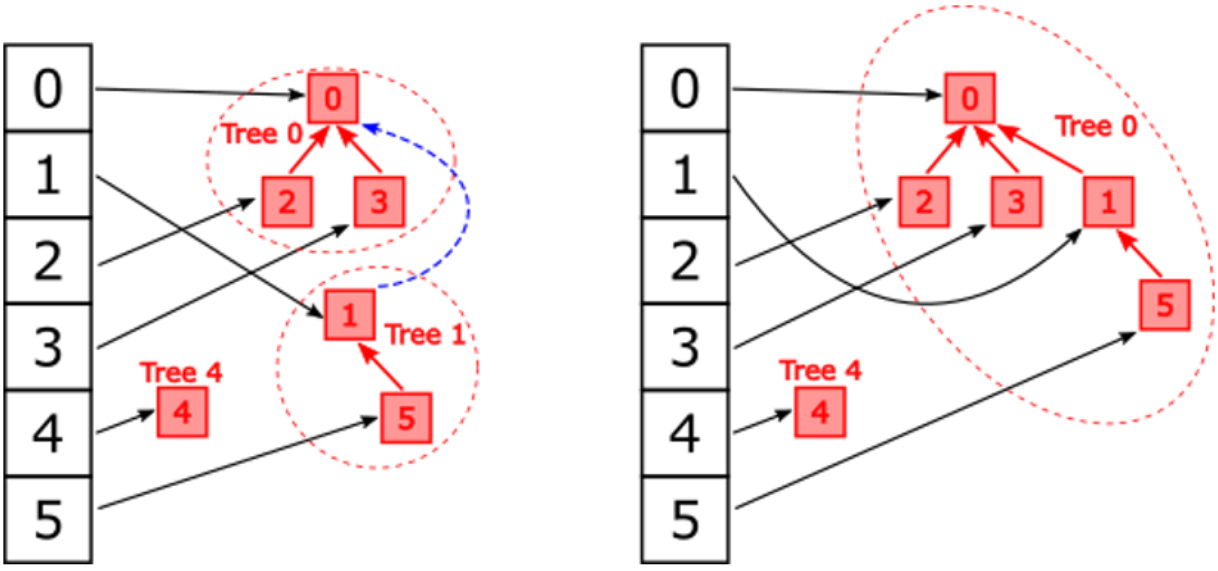


Figure 5.8 Merging two sets when using the tree representation: it only requires creating one new link (plus some tree traversing). In the figure on the left,